

School ERP Project

MySQL + Microsoft SQL Server (MSSQL) Integration Guide

Project Purpose

The School ERP system uses:

1. **MySQL** (Laravel Application Database)
2. Stores new ERP data.
3. Users, roles, permissions, exams, results, subjects, standards, divisions, etc.
4. **Microsoft SQL Server (Old ERP Database)**
5. Stores legacy student information.
6. Student records, roll numbers, admissions, divisions, parent details, etc.

Laravel connects to both databases simultaneously.

1. Required Software

A. Laravel Development

Install:

PHP 8.3+

Download: <https://windows.php.net/download>

Composer

Download: <https://getcomposer.org>

Verify:

```
php -v  
composer -V
```

B. MySQL Database

Install either:

Option 1 (Recommended)

XAMPP

Download: <https://www.apachefriends.org>

Contains:

- Apache
 - PHP
 - MySQL (MariaDB)
-

Option 2

MySQL Community Server

Download:

<https://dev.mysql.com/downloads/mysql>

C. Microsoft SQL Server

Install:

SQL Server Express

Download:

<https://www.microsoft.com/en-us/sql-server/sql-server-downloads>

Install:

- Database Engine
-

SQL Server Management Studio (SSMS)

Download:

<https://learn.microsoft.com/sql/ssms/download-sql-server-management-studio-ssms>

Used for:

- Viewing tables
- Running queries
- Database backup/restore

2. MSSQL Driver Installation for PHP

Laravel cannot connect to SQL Server until Microsoft drivers are installed.

Step 1

Install Microsoft ODBC Driver

Download:

<https://learn.microsoft.com/sql/connect/odbc/download-odbc-driver-for-sql-server>

Install:

ODBC Driver 18 for SQL Server

Step 2

Install Microsoft PHP SQL Server Drivers

Download:

<https://github.com/microsoft/msphpsql/releases>

Example:

```
php_sqlsrv_83_ts_x64.dll  
php_pdo_sqlsrv_83_ts_x64.dll
```

(Version depends on your PHP version.)

Step 3

Copy DLL files to:

```
PHP/ext/
```

Example:

```
E:\php\ext
```

or

```
C:\xampp\php\ext
```

Step 4

Edit php.ini

Add:

```
extension=sqlsrv  
extension=pdo_sqlsrv
```

Restart Apache.

Step 5

Verify Installation

Run:

```
php -m
```

You should see:

```
pdo_sqlsrv
sqlsrv
```

3. Laravel Database Configuration

File:

```
project/.env
```

MySQL Connection

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=school_management
DB_USERNAME=root
DB_PASSWORD=
```

MSSQL Connection

Add:

```
OLDERP_DB_HOST=192.168.1.100
OLDERP_DB_PORT=1433
OLDERP_DB_DATABASE=SchoolERP
OLDERP_DB_USERNAME=sa
OLDERP_DB_PASSWORD=password
```

4. Configure config/database.php

Add second connection:

```
'sqlsrv_olderp' => [  
    'driver' => 'sqlsrv',  
    'host' => env('OLDERP_DB_HOST'),  
    'port' => env('OLDERP_DB_PORT'),  
    'database' => env('OLDERP_DB_DATABASE'),  
    'username' => env('OLDERP_DB_USERNAME'),  
    'password' => env('OLDERP_DB_PASSWORD'),  
    'charset' => 'utf8',  
    'prefix' => '',  
],
```

5. Using MySQL Database

Default Laravel queries:

```
$students = Student::all();
```

or

```
DB::table('students')->get();
```

Uses:

```
DB_CONNECTION=mysql
```

automatically.

6. Using MSSQL Database

Example:

```
DB::connection('sqlsrv_olderp')  
->table('FeeMstStudent')  
->get();
```

Example:

```
$students =  
DB::connection('sqlsrv_olderp')  
->table('SubStudentMst')  
->get();
```

7. School ERP Actual Usage

MySQL Tables

Examples:

```
users  
roles  
permissions  
academic_years  
standards  
divisions  
subjects  
subject_types  
exam_masters  
student_marks  
student_results  
student_result_details  
teacher_class_allocations  
teacher_subject_allocations  
student_skill_subjects
```

MSSQL Tables

Examples:

```
FeeMstStudent  
SubStudentMst  
StandardMst  
DivisionMst
```

Used for:

- Student Name
- Roll Number
- Parent Mobile
- Admission Information

8. Example Helper Method

Current School ERP implementation:

```
DB::connection('sqlsrv_olderp')  
->table('SubStudentMst as s')  
->join(  
    'FeeMstStudent as f',  
    'f.Studentid',  
    '=',  
    's.Studentid'  
)  
->where('s.yearid', $yearId)  
->get();
```

This reads student data from MSSQL while results are stored in MySQL.

9. Testing Connections

MySQL

```
Route::get('/mysql-test', function () {
```



```
return DB::connection('mysql')
    ->select('SELECT 1');
});
```

MSSQL

```
Route::get('/mssql-test', function () {

    return DB::connection('sqlsrv_olderp')
        ->select('SELECT TOP 1 * FROM FeeMstStudent');
});
```

10. Common Errors

Error

```
could not find driver
```

Solution:

Install:

```
pdo_sqlsrv
sqlsrv
```

extensions.

Error

```
Login failed for user
```

Solution:

Verify:

OLDERP_DB_USERNAME
OLDERP_DB_PASSWORD

Error

SQLSTATE[08001]

Solution:

Check:

- SQL Server service running
 - TCP/IP enabled
 - Firewall open
 - Port 1433 reachable
-

11. Verify SQL Server TCP/IP

Open:

SQL Server Configuration Manager

Navigate:

SQL Server Network Configuration

Enable:

TCP/IP

Restart:

SQL Server Service

Current School ERP Architecture

```
Laravel 13
|
├── MySQL
|   ├── Users
|   ├── Exams
|   ├── Marks
|   ├── Results
|   └── Master Data
└── SQL Server (Old ERP)
    ├── Students
    ├── Roll Numbers
    ├── Standards
    ├── Divisions
    └── Parent Information
```

This architecture allows the School ERP to continue using student data from the old ERP while storing all new ERP functionality in MySQL.